Lesson 03 Demo 01

Deploying MySQL Database as Docker Container

Objective: To demonstrate how to deploy a MySQL database as a Docker container on an AWS EC2 instance with Ubuntu

Tools required: AWS Management Console

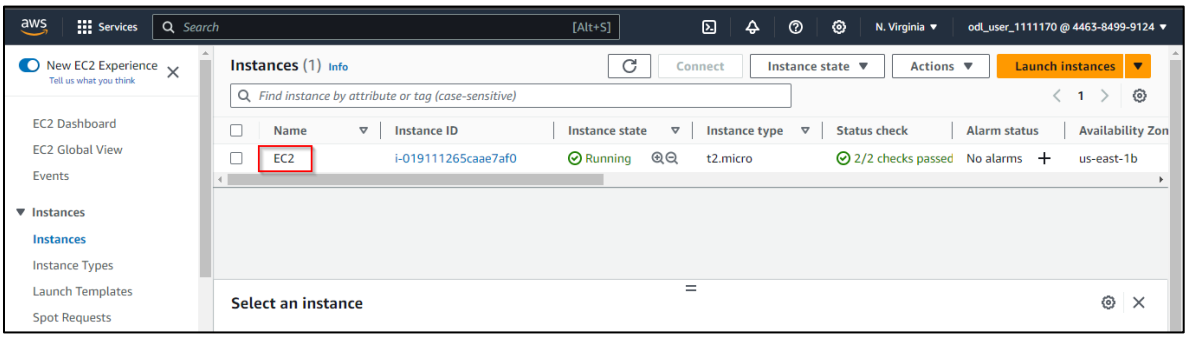
Prerequisites: An AWS EC2 instance with Ubuntu as the operating system

Steps to be followed:

1. Set up an EC2 instance
2. Install the Docker on Ubuntu
3. Run the MySQL container
4. Connect to MySQL server within container

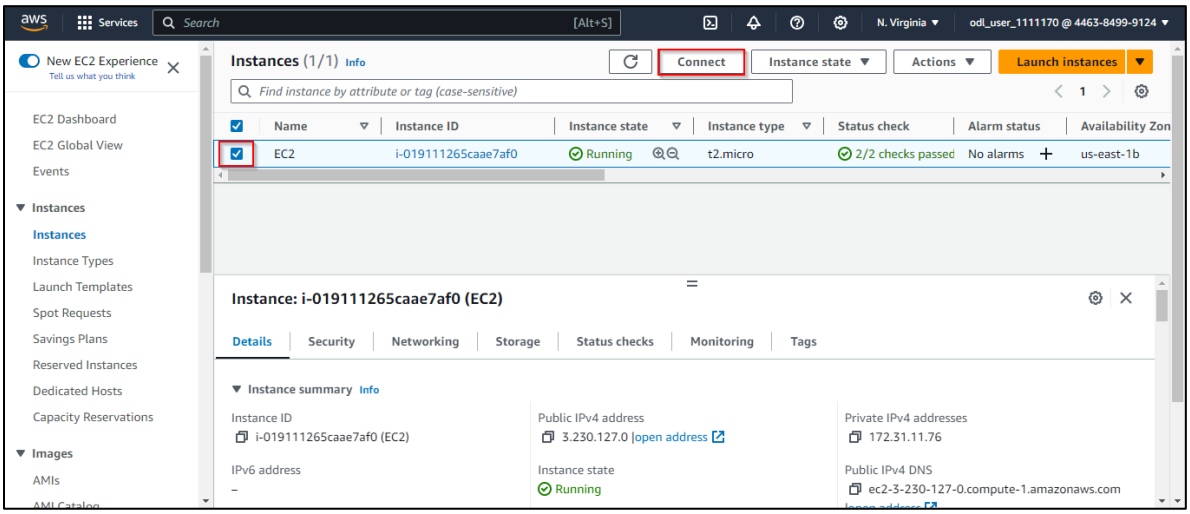
Step 1: Set up an EC2 instance

1.1 Launch a new EC2 instance with Ubuntu as the operating system. Ensure that you have the necessary security group rules to allow SSH access.

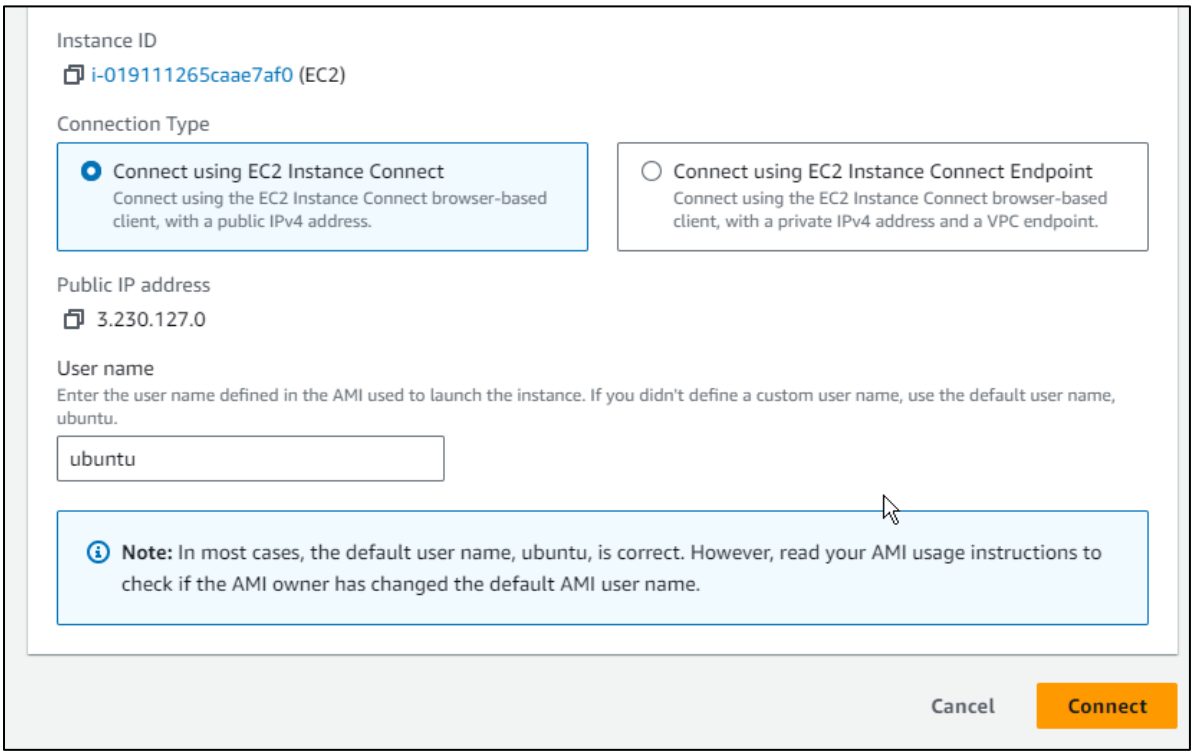


Note: Please refer to previous lesson demos on how to launch an EC2 instance.

1.2 Select the instance and click **Connect**



1.3 Click **Connect** again to proceed



Step 2: Install the Docker on Ubuntu

2.1 Update the package list by running the following commands:

sudo su

apt update

```
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
ubuntu@ip-172-31-11-76:~$ apt get update
E: Invalid operation get
ubuntu@ip-172-31-11-76:~$ apt update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
ubuntu@ip-172-31-11-76:~$ sudo su
root@ip-172-31-11-76:/home/ubuntu# apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu focal InRelease
Hit:6 https://packages.cloud.google.com/apt kubernetes-xenial InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
106 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-11-76:/home/ubuntu#
```

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2.2 Run the following command to install the necessary dependencies:

apt install apt-transport-https ca-certificates curl software-properties-common -y

```
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt,it does not contain exactly one certificate or CRL
19 added, 6 removed; done.
Setting up python3-software-properties (0.99.9.12) ...
Setting up software-properties-common (0.99.9.12) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for dbus (1.12.16-2ubuntu2.3) ...
Processing triggers for ca-certificates (20230311ubuntu0.20.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@ip-172-31-11-76:/home/ubuntu# apt install apt-transport-https ca-certificates curl software-properties-common -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
ca-certificates is already the newest version (20230311ubuntu0.20.04.1).
ca-certificates set to manually installed.
curl is already the newest version (7.68.0-1ubuntu2.19).
software-properties-common is already the newest version (0.99.9.12).
apt-transport-https is already the newest version (2.0.9).
0 upgraded, 0 newly installed, 0 to remove and 103 not upgraded.
root@ip-172-31-11-76:/home/ubuntu#
```

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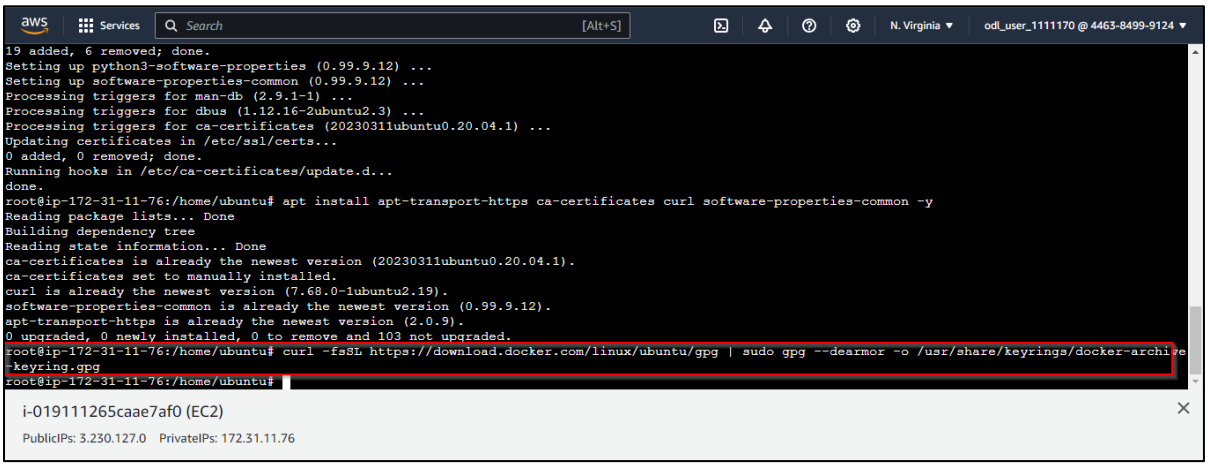
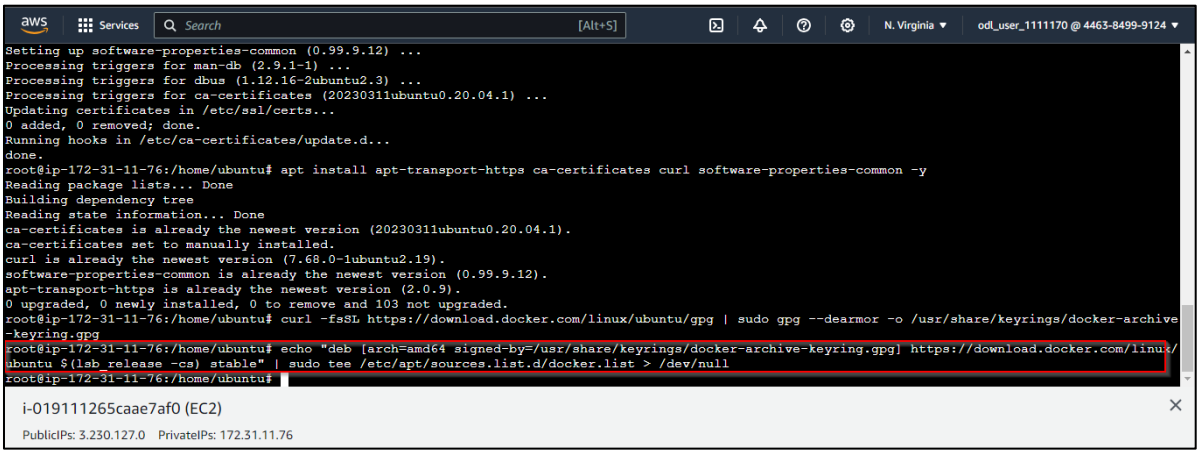
2.3 Run the following command to add the Docker GPG key:

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg



2.4 Run the following command to add the Docker repository:

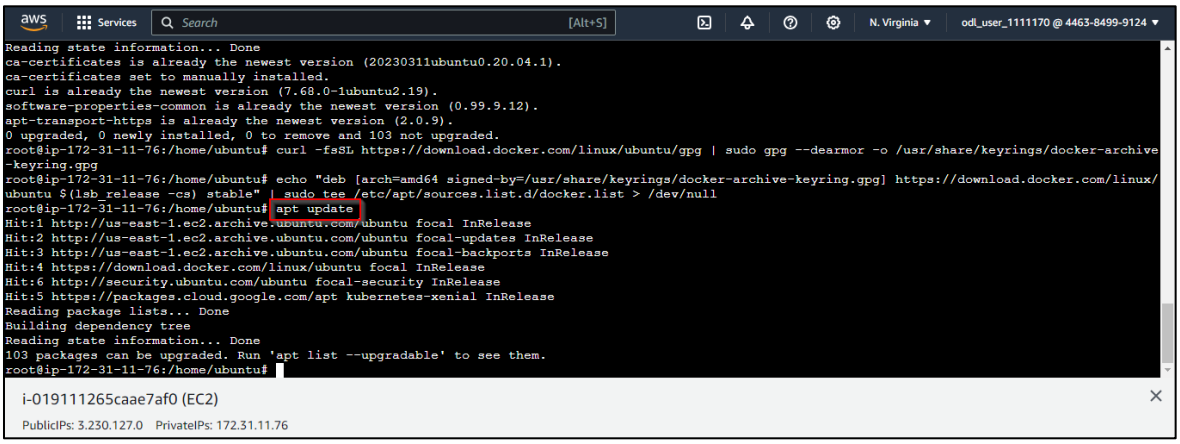
echo "deb [arch=amd64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

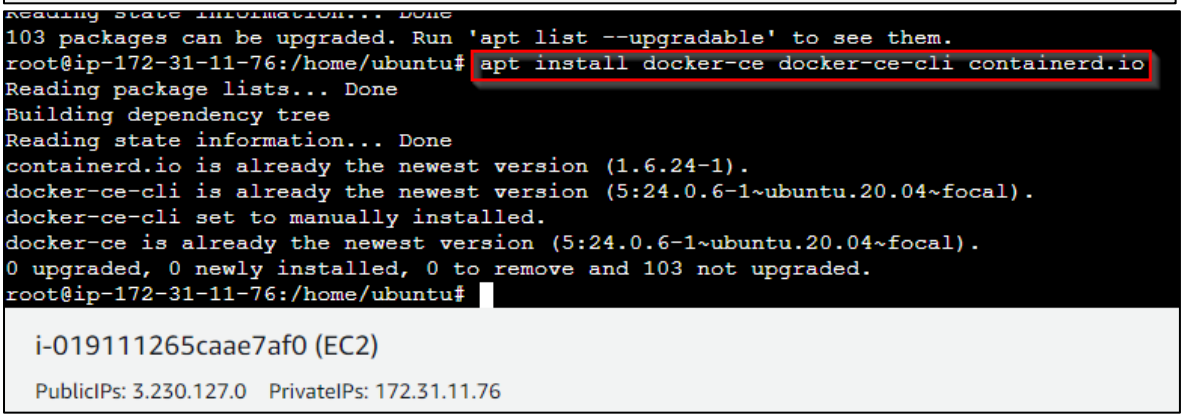


2.5 Run the following commands to install Docker:

apt update

apt install docker-ce docker-ce-cli containerd.io





2.6 Run the following commands to start and enable the Docker service:

systemctl start docker

systemctl enable docker

```
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 https://download.docker.com/linux/ubuntu focal InRelease
Hit:6 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:5 https://packages.cloud.google.com/apt kubernetes-xenial InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
103 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-11-76:/home/ubuntu# apt install docker-ce docker-ce-cli containerd.io
Reading package lists... Done
Building dependency tree
Reading state information... Done
containerd.io is already the newest version (1.6.24-1).
docker-ce-cli is already the newest version (5:24.0.6-1~ubuntu.20.04~focal).
docker-ce-cli set to manually installed.
docker-ce is already the newest version (5:24.0.6-1~ubuntu.20.04~focal).
0 upgraded, 0 newly installed, 0 to remove and 103 not upgraded.
root@ip-172-31-11-76:/home/ubuntu# systemctl start docker
root@ip-172-31-11-76:/home/ubuntu# systemctl enable docker
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
root@ip-172-31-11-76:/home/ubuntu#
```

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Step 3: Run the MySQL container

3.1 Pull the latest MySQL image from Docker Hub using the following command:

docker pull mysql

```
docker-ce is already the newest version (5:24.0.6-1~ubuntu.20.04~focal).
0 upgraded, 0 newly installed, 0 to remove and 103 not upgraded.
root@ip-172-31-11-76:/home/ubuntu# systemctl start docker
root@ip-172-31-11-76:/home/ubuntu# systemctl enable docker
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
root@ip-172-31-11-76:/home/ubuntu# docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
5262579e8e45: Pull complete
741b767e25b7: Pull complete
06e0c37837cf: Pull complete
c6f5d3670db7: Pull complete
d5c567b29c3e: Pull complete
6effcc6561c9: Pull complete
1e1493d45d9c: Pull complete
7101609fa7d9: Pull complete
432a1261dc2a: Pull complete
865a24d6d1f2: Pull complete
Digest: sha256:566007208a3f1cc8f9df6b767665b5c9b800fc4fb5f863d17aa1df362880ed04
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
root@ip-172-31-11-76:/home/ubuntu#
```

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3.2 Create a Docker volume or bind mount to store your database data persistently. For example, create a volume named **mysql-data**:
docker volume create mysql-data

```
c6f5d3670db7: Pull complete
d5c567b29c3e: Pull complete
6effcc6561c9: Pull complete
1e1493d45d9c: Pull complete
7101609fa7d9: Pull complete
432a1261dc2a: Pull complete
865a24d6d1f2: Pull complete
Digest: sha256:566007208a3f1cc8f9df6b767665b5c9b800fc4fb5f863d17aa1df362880ed04
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
root@ip-172-31-11-76:/home/ubuntu# docker volume create mysql-data
mysql-data
root@ip-172-31-11-76:/home/ubuntu#
```

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3.3 Run a MySQL container and set a password for the root user using the **MYSQL_ROOT_PASSWORD** environment variable:
docker run --name my-mysql -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -d mysql

aws Services Search [Alt+S] N. Virginia odl_user_111

```
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
root@ip-172-31-11-76:/home/ubuntu# docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
5262579e8e45: Pull complete
741b767e25b7: Pull complete
06e0c37837cf: Pull complete
c6f5d3670db7: Pull complete
d5c567b29c3e: Pull complete
6effcc6561c9: Pull complete
1e1493d45d9c: Pull complete
7101609fa7d9: Pull complete
432a1261dc2a: Pull complete
865a24d6d1f2: Pull complete
Digest: sha256:566007208a3f1cc8f9df6b767665b5c9b800fc4fb5f863d17aa1df362880ed04
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
root@ip-172-31-11-76:/home/ubuntu# docker volume create mysql-data
mysql-data
root@ip-172-31-11-76:/home/ubuntu# docker run --name my-mysql -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -d mysql
5414866cef442ec61b8543748d905a0dcca181a74483b8672cbe5ba5eab44cf8
root@ip-172-31-11-76:/home/ubuntu#
```

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3.4 Check the status of your container using the command:
docker ps

```
1e1493d45d9c: Pull complete
7101609fa7d9: Pull complete
432a1261dc2a: Pull complete
865a24d6d1f2: Pull complete
Digest: sha256:566007208a3f1cc8f9df6b767665b5c9b800fc4fb5f863d17aa1df362880ed04
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
root@ip-172-31-11-76:/home/ubuntu# docker volume create mysql-data
mysql-data
root@ip-172-31-11-76:/home/ubuntu# docker run --name my-mysql -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -d mysql
5414866cef442ec61b8543748d905a0dcca181a74483b8672cbe5ba5eab44cf8
root@ip-172-31-11-76:/home/ubuntu# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS          PORTS                   NAMES
5414866cef44   mysql     "docker-entrypoint.s…"   About a minute ago   Up 59 seconds   3306/tcp, 33060/tcp     my-mysql
root@ip-172-31-11-76:/home/ubuntu#
```

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Step 4: Connect to MySQL server within container

4.1 Connect to your MySQL server from within the container:
docker exec -it my-mysql mysql -p

```
root@ip-172-31-11-76:/home/ubuntu# docker volume create mysql-data
mysql-data
root@ip-172-31-11-76:/home/ubuntu# docker run --name my-mysql -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -d mysql
5414866cef442ec61b8543748d905a0dcca181a74483b8672cbe5ba5eab44cf8
root@ip-172-31-11-76:/home/ubuntu# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS          PORTS                   NAMES
5414866cef44   mysql     "docker-entrypoint.s…"   About a minute ago   Up 59 seconds   3306/tcp, 33060/tcp     my-mysql
root@ip-172-31-11-76:/home/ubuntu# docker exec -it my-mysql mysql -p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
root@ip-172-31-11-76:/home/ubuntu# docker exec -it my-mysql mysql -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.1.0 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

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Note: Use **secret** as the password

4.2 Use the MySQL command-line interface to create databases, tables, and users and to perform queries. For example, create a database named **testdb**:
CREATE DATABASE testdb;

```
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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.01 sec)

mysql>
```

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4.3 Exit the MySQL shell using the command:
exit

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.1.0 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.01 sec)

mysql> exit
Bye
root@ip-172-31-11-76:/home/ubuntu#
```

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4.4 Expose container port 3306 to host port 33060 using the following commands:

docker run --name my-mysql1 -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -p 33060:3306 -d mysql

docker ps -a

```
mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.01 sec)

mysql> exit
Bye
root@ip-172-31-11-76:/home/ubuntu# docker run --name my-mysql -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -p 33060:3306 -d mysql
docker: Error response from daemon: Conflict. The container name "/my-mysql" is already in use by container "5414866cef442ec61b8543748d905a0dcca181a
4483b8672cbe5ba5eab44cf8". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
root@ip-172-31-11-76:/home/ubuntu# docker run --name my-mysql1 -e MYSQL_ROOT_PASSWORD=secret -v mysql-data:/var/lib/mysql -d mysql
ded8be56bb7794eb1c6bc3d9158da321ed4cd782801c8621252582becd081853
root@ip-172-31-11-76:/home/ubuntu# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                   NAMES
ded8be56bb77   mysql     "docker-entrypoint.s…"   57 seconds ago   Up 56 seconds   3306/tcp, 33060/tcp     my-mysql1
5414866cef44   mysql     "docker-entrypoint.s…"   9 minutes ago    Up 9 minutes    3306/tcp, 33060/tcp     my-mysql
root@ip-172-31-11-76:/home/ubuntu#
```

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By following these steps, you have successfully deployed a MySQL database as a Docker container on your AWS EC2 instance running Ubuntu.